



From 90-Day Sprint to Sustained Scale: Engineering Trust into Gov 4.0's Agentic AI

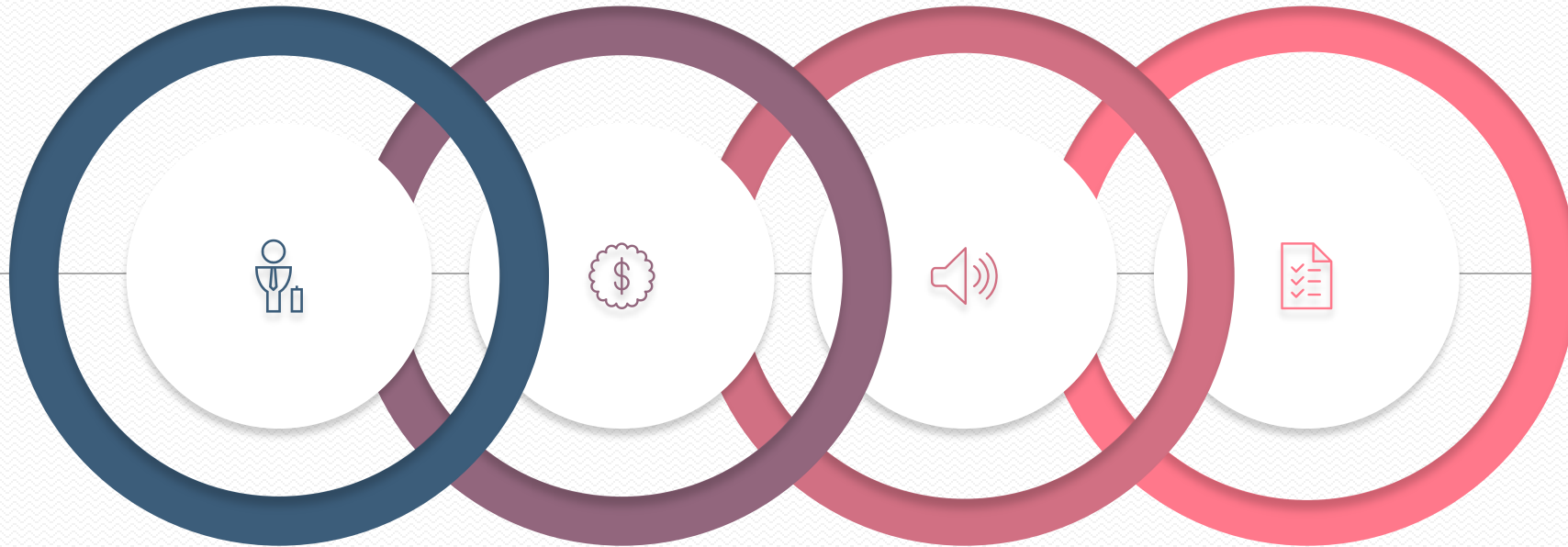
ABOUT ME

ANITHA SENTHILANTHAN

AI & Cloud Solutions Architect

- 13+ years of application development and architecting enterprise-grade systems that run inside real business operations.
- Multi-cloud certified (AWS, Azure, GCP).
- Focused on one problem: making AI agents fast and trustworthy at enterprise scale.
- AWS Community Builder (Security).

The Enterprise AI Reality Check

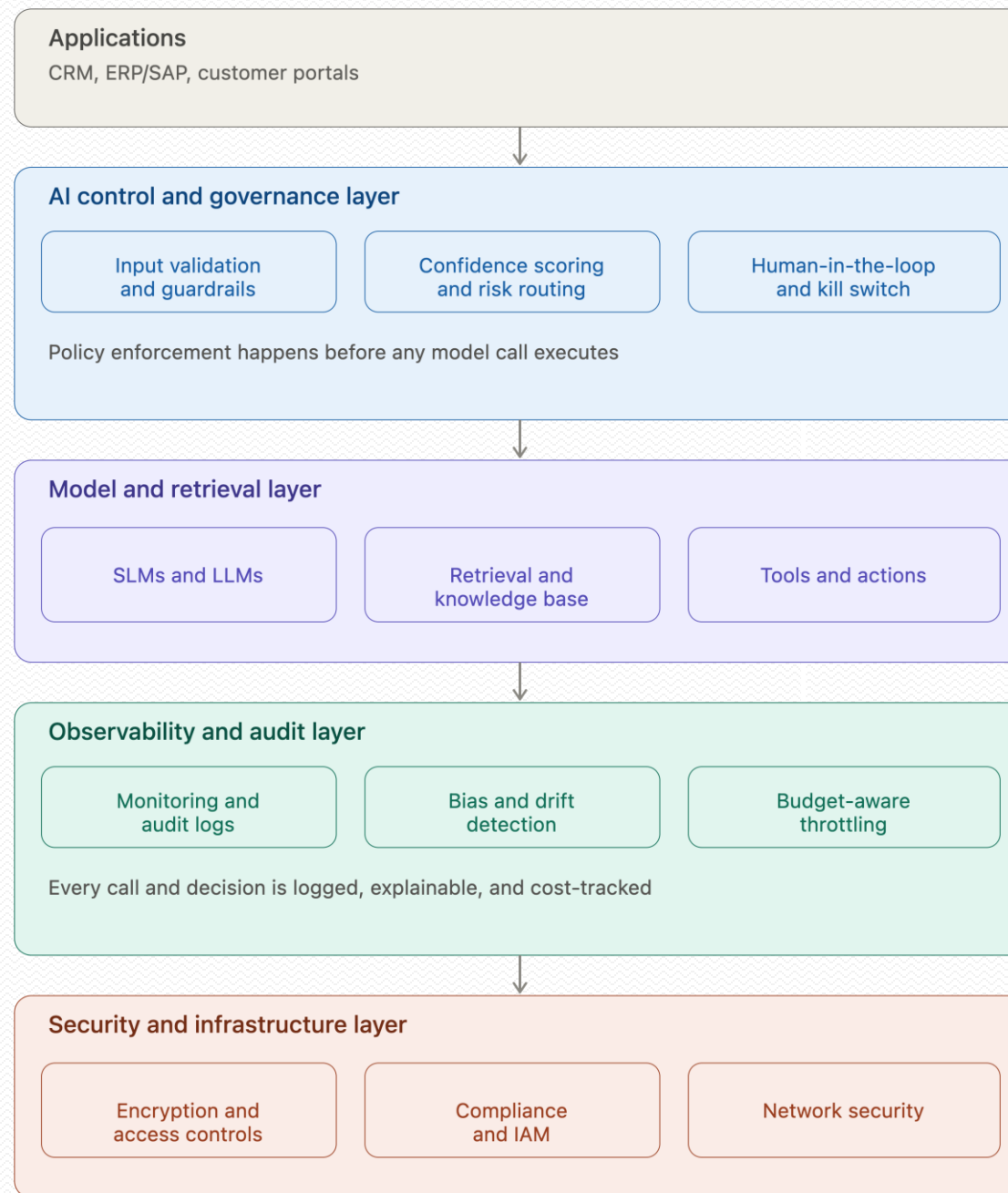


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Agents move faster than humans and fail faster too.

● ●
Trust failures: hallucinations, wrong actions, data leakage, no audit trail.

● ● ●
Cost failures: runaway token usage, duplicate tool calls, infrastructure sprawl.

● ● ● ●
At scale, trust and cost stop being engineering concerns; they become board-level metrics.



How We Monitor AI Agents Safely at Scale

Testing Accuracy



Validate expected answers
Run test datasets
Compare against ground truth

Bias & Fairness Layer



Bias testing
Fairness measurement
Segment-level outcome checks
Measure confidence score
Detect weak grounding
Calculate hallucination risk

Confidence & Hallucination Score



Adaptive Learning Loop



Capture feedback
Update prompts, tools, and retrieval
Retrain or fine-tune only with governance approval

Model Drift Monitoring



Monitor data drift
Monitor response quality
Detect behavior changes over time

Practical AI Agent Scoring: From Answer to Evidence

User asks

“Show medium revenue license deals”

Answer: 18 open license opportunities, AED 6.2M total pipeline

Confidence: 92%

Evidence: SQL rule + data freshness + validation status

Validation checks

Intent matched: License opportunity query

Business rule matched: Revenue between AED 100K–500K

SQL filter validated: Opportunity type, status, revenue band

Data completeness checked: Revenue, stage, close date, owner

Freshness checked: CRM data updated within SLA

Confidence Score = Weighted Reliability Score

30% Intent Confidence

25% Data Completeness

20% Rule Match Accuracy

15% Retrieval / SQL Accuracy

10% Freshness Score

Production Monitoring: Drift, Bias, Fairness, and Adaptive Learning

1. Model Drift Monitoring

Track intent accuracy over time
Monitor SQL failure rate
Detect new CRM fields, stages, regions, opportunity types
Watch confidence score trend and user correction rate

Example

Last month: intent accuracy 96%
This month: intent accuracy 84%
Root cause: new opportunity type — “Subscription License”

2. Bias & Fairness Layer

Compare recommendation rates by region, owner, segment, industry, deal size
Detect skew in recommendations
Flag unfair prioritization before business decisions

Example

CRM pipeline: UAE 40%, KSA 35%
Agent recommendations: UAE 80%, KSA 15%

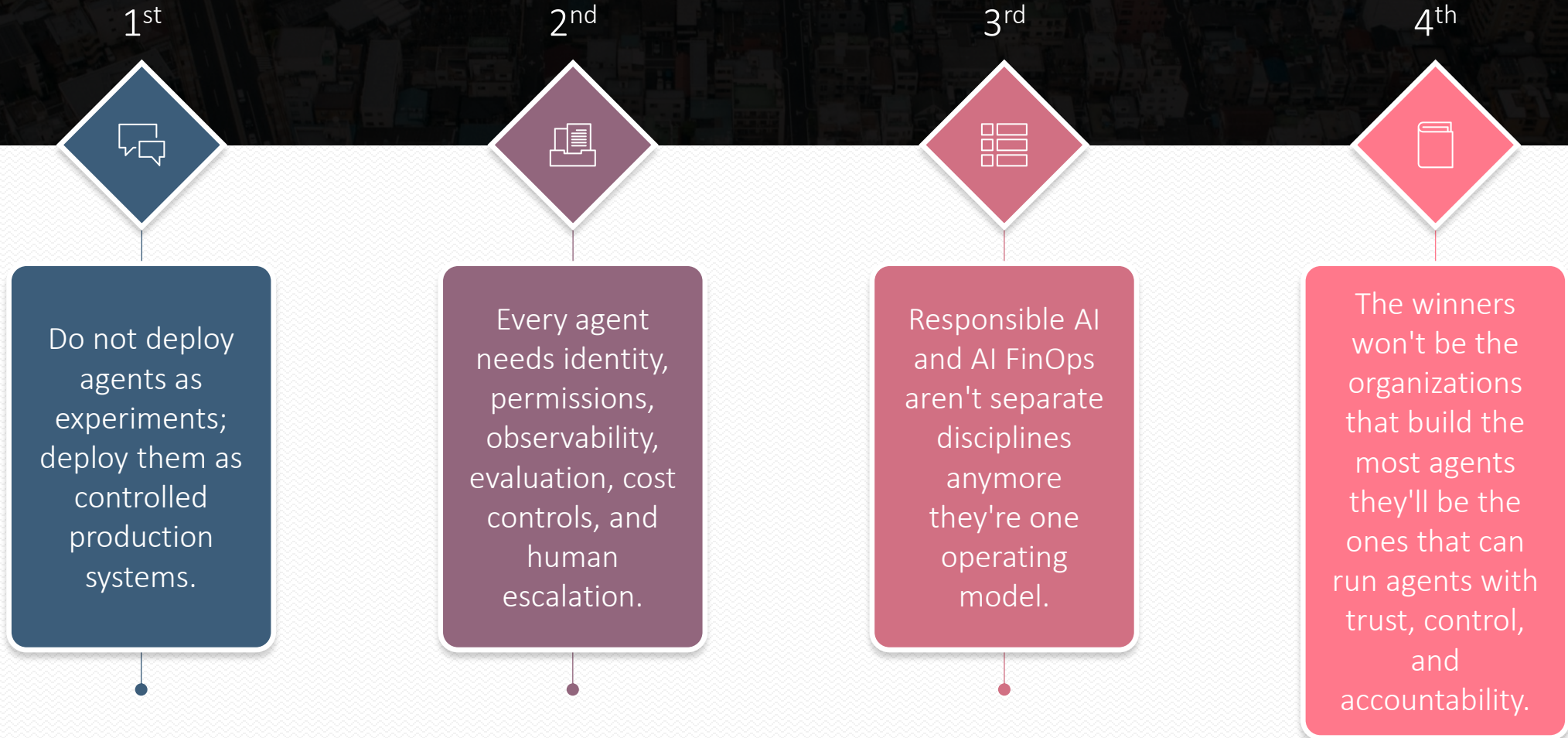
3. Governed Adaptive Learning

Capture user corrections
Update glossary, rules, prompts, index, and test cases
Re-run evaluation
Deploy through controlled release

Example

Correction logged → rules and tests updated → evaluation re-run → controlled release

Scale Agents Like Critical Enterprise Systems



Let's Test Your Governance Pulse!

A Revenue Agent gives this answer:

“Next month best-case pipeline is AED 5M with 91% confidence.”

During audit, you find:

- The SQL query was correct.
- The CRM data was fresh.
- The agent retrieved only authorized records.
- But the agent consistently ignored smaller high-probability deals and mostly recommended large enterprise accounts.

- A. Hallucination
- B. Model drift
- C. Fairness / business bias issue
- D. Access control failure

What is the most accurate issue?

Correct Answer:  C) Fairness / business bias issue

Let's Test Your Governance Pulse!

A procurement AI agent recommends approving a supplier contract. The confidence score is **88%**, the data is fresh, and the supplier risk score is low. However, the contract value is above the company's approval threshold.

What should the system do?

- A. Approve automatically because confidence is high.
- B. Reject automatically because the contract value is high.
- C. Escalate to a human approver because it is a high-impact action.
- D. Ask the AI model to generate a second opinion

Correct Answer:  C) Escalate to a human approver because it is a high-impact action.



Thank You.

Anitha Senthilnathan